

Breast Cancer Tumor Classification



Using Clinical Features & Machine Learning

Logistic Regression

SVM (RBF Kernel)

Random Forest

SHAP Explainability



Problem Statement

The Challenge

Accurate classification of breast tumors as Malignant or Benign from digitized cell nuclei images.

Clinical Impact

Early detection of malignant tumors is critical; it directly improves patient survival rates and guides treatment decisions.

Objective

Build & compare ML models achieving >95% accuracy, with high recall to minimize missed cancer cases (False Negatives).

Dataset at a Glance

568

Total Samples

30

Feature Columns

357

Benign (63%)

211

Malignant (37%)

0

Missing Values



Dataset Overview

Mean Values

- radius_mean
- texture_mean
- perimeter_mean
- area_mean
- smoothness_mean
- compactness_mean

Standard Error (SE)

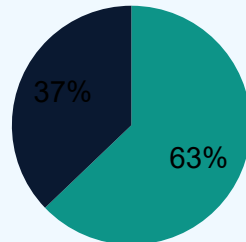
- radius_se
- texture_se
- perimeter_se
- area_se
- smoothness_se
- compactness_se

Worst Values

- radius_worst
- texture_worst
- perimeter_worst
- area_worst
- concave_points_worst
- concavity_worst

Target Variable: outcome

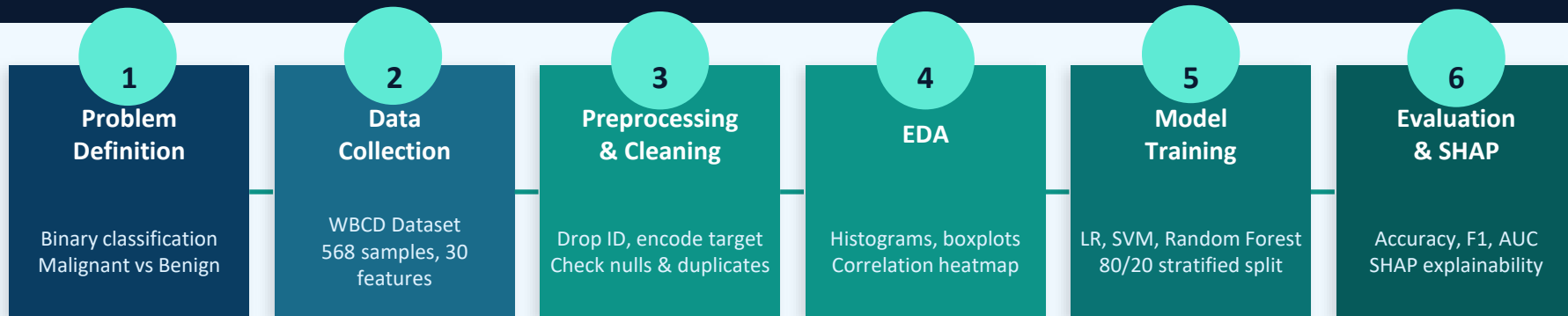
M = Malignant (1) | B = Benign (0)



- Benign (63%)
- Malignant (37%)



Methodology & Pipeline



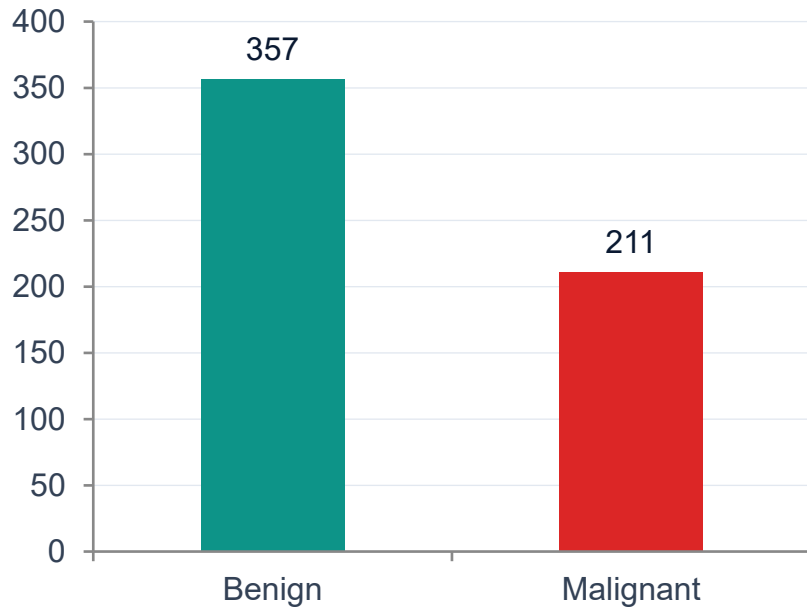
Key Techniques Applied

- ✓ StandardScaler for feature normalization
- ✓ Stratified 80/20 train-test split
- ✓ 5-Fold Stratified Cross-Validation
- ✓ SHAP & Permutation Importance for model explainability
- ✓ Confusion Matrices & ROC-AUC for evaluation
- ✓ joblib model serialization (.pkl files)



Exploratory Data Analysis

Class Distribution



Key EDA Findings



Class Balance:

Dataset is slightly imbalanced; 63% Benign vs 37% Malignant. Suitable for stratified splitting.



Feature Distributions:

Most features show normal/skewed distributions. Histograms reveal overlap between classes.



Outliers Detected:

Boxplots reveal outliers in several features; not extreme enough to impact analysis significantly.



High Correlations:

Radius, Perimeter & Area are strongly correlated (redundancy). Concave features are key discriminators.



Clear Separation:

Scatter plot of radius_mean vs area_mean shows clear visual separation between Malignant & Benign.

Model Building & Training

Logistic Regression

Parameters

```
max_iter=1000  
random_state=42
```

Advantages

- ✓ Simple & interpretable
- ✓ Fast training
- ✓ Good baseline

SVM (RBF Kernel)

Parameters

```
kernel='rbf'  
probability=True
```

Advantages

- ✓ Handles non-linearity
- ✓ Robust to outliers
- ✓ Strong generalization

Random Forest

Parameters

```
n_estimators=100  
random_state=42
```

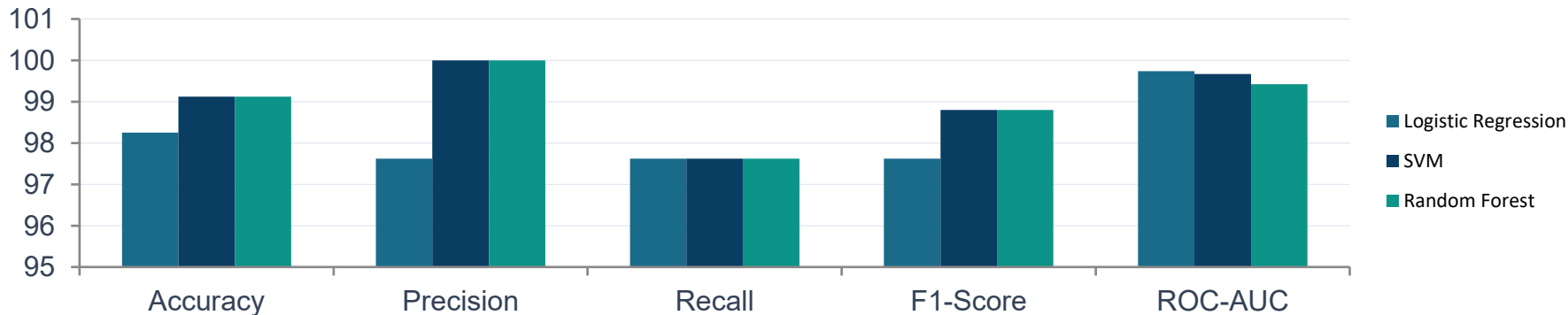
Advantages

- ✓ Ensemble method
- ✓ Feature importance
- ✓ Handles redundancy



Key Results & Model Comparison

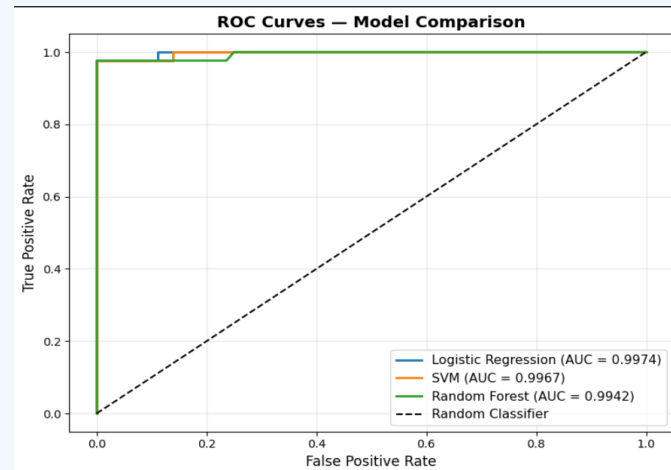
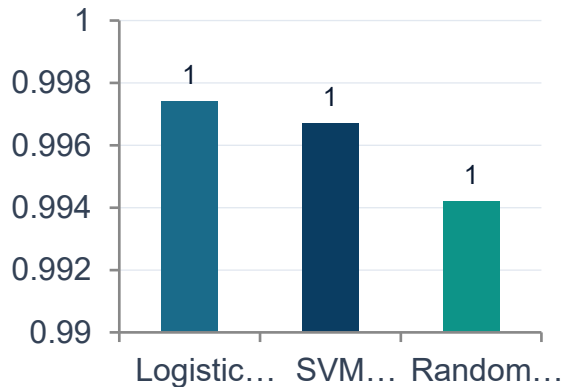
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	98.25%	97.62%	97.62%	97.62%	99.74%
SVM (RBF Kernel)	99.12%	100.00%	97.62%	98.80%	99.67%
Random Forest	99.12%	100.00%	97.62%	98.80%	99.42%



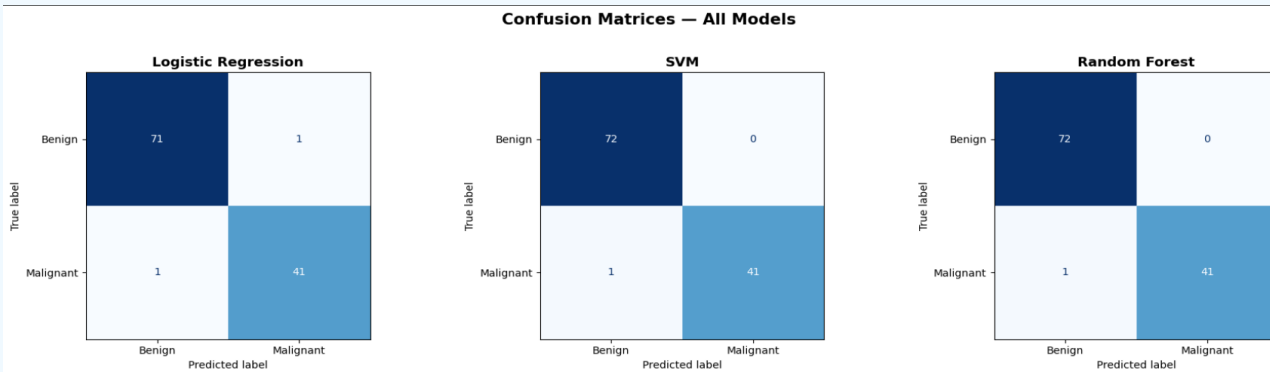
All three models exceed 98% accuracy — SVM & Random Forest tie at 99.12%

ROC Curves & Evaluation Charts

ROC-AUC Comparison



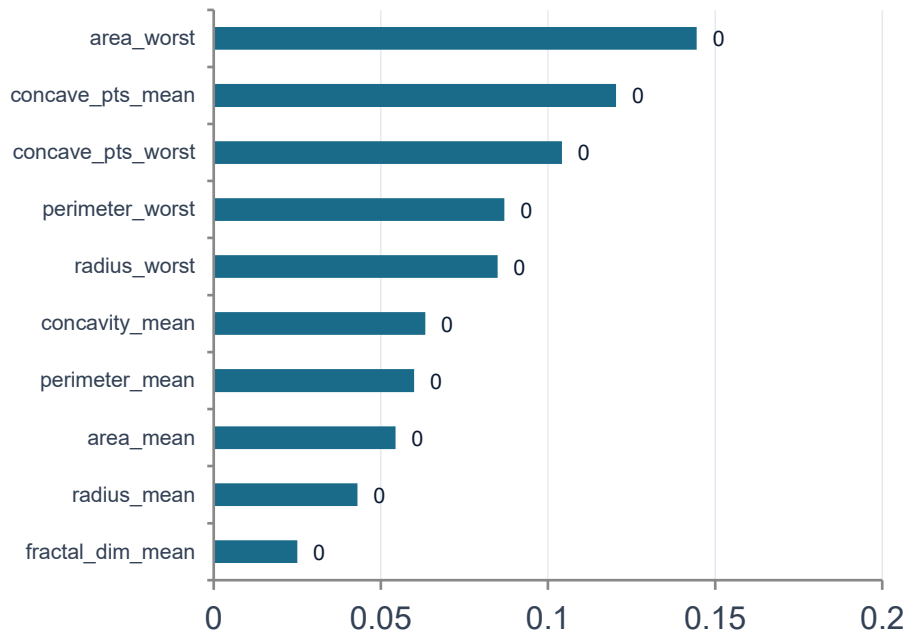
Confusion Matrices — All Models





Feature Importance & SHAP Explainability

Random Forest-Top 10 Features



SHAP Explainability Methods

RF Feature Importance

Global feature ranking by Gini impurity

LR Coefficients

Direction & magnitude of each feature's influence

Permutation Importance

True impact on F1 score per model

SHAP Summary (Dot)

Feature impact distribution across samples

SHAP Summary (Bar)

Mean absolute SHAP values (global importance)

SHAP Waterfall

Single prediction explained feature-by-feature

Key Finding: area_worst, concave_points_mean & concave_points_worst are the most consistent predictors across all models.



Deployed Application

Course: Predictive Analytics | Heart disease prediction with n | Breast Cancer Tumor Classifier | Breast Cancer Tumor Classifier | +

localhost:8501

Settings

Choose Model

Logistic Regression

Batch Prediction (CSV)

Upload a CSV with 30 features

Upload

200MB per file • CSV

Medical Disclaimer: This tool is for educational and research purposes only. It is not a substitute for professional medical diagnosis. Always consult a qualified clinician.

Manual Input | Batch CSV | Model Info

Enter Cell Nucleus Measurements

Mean Values – average of cell nuclei | SE Values – standard error of measurements | Worst Values – largest/worst measurements

Mean Values

Radius Mean	Texture Mean
13.37000	19.28000
Perimeter Mean	Area Mean
86.24000	551.10000
Smoothness Mean	Compactness Mean
0.09600	0.10400

Model Performance Summary

Model	Accuracy	ROC-AUC
Logistic Reg	0.9825	0.9974
SVM	0.9912	0.9967
Random Forest	0.9912	0.9942

Breast Cancer Tumor Classifier | +

breast-cancer-tumor-classification-5pqgbbqfscd0lleuwlaustreamlit.app

Choose Model

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Prediction Result – Logistic Regression

BENIGN

The model predicts the tumor is benign (non-cancerous).

Confidence: 67.8%

Regular monitoring is still recommended.

Class Probabilities

Class	Probability (%)
Malignant	32.2%
Benign	67.8%

All Models Comparison

Model	Confidence
Logistic Regression	Conf: 67.8%
SVM (RBF Kernel)	Conf: 92.7%
Random Forest	Conf: 72.0%

Model Performance Summary

Model	Accuracy	ROC-AUC
Logistic Reg	0.9825	0.9974
SVM	0.9912	0.9967
Random Forest	0.9912	0.9942

Evaluated on 20% held-out test set.

Breast Cancer Tumor Classifier - Digital University Kerala - Predictive Analytics 2025-26 - Built with Streamlit

Manage app

Conclusions, Limitations & Future Work



Conclusions

- ▶ All 3 models achieved >98% accuracy on the Wisconsin Breast Cancer Dataset.
- ▶ SVM & Random Forest tied at 99.12% accuracy with 100% Precision.
- ▶ Logistic Regression achieved the highest ROC-AUC (0.9974); best for screening.
- ▶ Key features: area_worst, concave_points_mean, perimeter_worst.
- ▶ SHAP analysis confirms consistent feature importance across all models.



Limitations

- ▶ Dataset is relatively small (568 samples); limited generalizability.
- ▶ Slight class imbalance (63% Benign vs 37% Malignant) may affect minority class.
- ▶ Features are pre-computed from images; raw image data not used.
- ▶ Models not validated on external clinical datasets.
- ▶ No hyperparameter tuning performed (Grid Search / Bayesian Optimization).



Future Work

- ▶ Apply deep learning (CNN) directly on cell nuclei images.
- ▶ Perform hyperparameter optimization with GridSearchCV.
- ▶ Handle class imbalance using SMOTE or class weights.
- ▶ Validate on real-world hospital datasets for clinical applicability.
- ▶ Integrate SHAP explanations into the deployed Streamlit app.